



Orbit determination, clock estimation and performance evaluation of BDS-3 PPP-B2b service

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Abstract

This paper focuses on PPP-B2b, one of the featured services for BDS-3, which provides users around China with centimetre-level static positioning accuracy and decimetre-level kinematic positioning accuracy by broadcasting precise corrections for GPS/BDS-3 satellites. The GEO PPP-B2b signal broadcasted information types, including the PRN mask, orbit corrections, differential code bias corrections and clock corrections, are introduced, as well as a brief description of their usage. A new orbit determination strategy using regional L-band code/phase measurements and inter-satellite link measurements in combination and a real-time clock estimation strategy based on the Kalman filter for PPP-B2b precise correction generation are introduced in this contribution. Then, the accuracy of the orbit and clock corrections is assessed. The orbit user ranging error (URE) is 0.05 m for the BDS-3 MEO and GPS satellites. The orbit URE for BDS-3 IGSO satellites is worse, i.e., 0.15 m. The clock correction accuracy is 0.2 ns for BDS-3 and GPS satellites. Finally, the PPP-B2b performance is validated by both the static PPP process and simulated kinematic PPP process. BDS-3/GPS dual-system PPP offers faster convergence and better accuracy. The positioning accuracy achievable using PPP-B2b real-time products is at the same level as that using post-processed products. The RMS for BDS-3/GPS dual-system static positioning errors is less than 1.0 cm in east and north and about 3.0 cm in the up. The simulated kinematic positioning accuracy is better than 2.5 cm in the north, 3.5 cm in the east and 8.5 cm in the up directions after convergence.

Highlights

- A new determination strategy using regional L-band code/phase measurements and inter-satellite link measurements in combination and a real-time clock estimation strategy based on the Kalman filter for PPP-B2b precise correction generation are introduced.
- User ranging error (URE) is 0.05 m for the BDS-3 MEO and GPS satellites.
- The clock correction accuracy is 0.2 ns for BDS-3 and GPS satellites.
- The positioning accuracy achievable using PPP-B2b real-time products is at the same level as that using post-processed products.

Keywords BeiDou navigation satellite system · Satellite orbits · Real-time clock estimation · Precise point positioning · Global positioning system

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1 Introduction

The BeiDou Navigation Satellite System (BDS), independently developed and operated by China, is regarded as a temporal-spatial infrastructure of national significance, providing all-time, all-weather and high-accuracy positioning, navigation and timing services to global users (CSNO 2019a). The BDS is listed as one of the four satellite navigation providers, along with the GPS of the US, GLONASS of Russia, and Galileo of Europe. The development of the BDS has followed a three-step strategy. The establishment of BDS-1, also named the experimental BeiDou navigation system, was finished in 2003, marking the end of the first step of BDS construction. Different from mainstream systems such as GPS and GLONASS, BDS-1 consists of only GEO satellites and provides a bidirectional interactive radio determination satellite service (RDSS). The second step of BDS construction, BDS-2, was finished by the end of 2012. BDS-2 consists of GEO, IGSO and MEO satellites and provides positioning, navigation and timing (PNT) services to users around the Asia–Pacific area (Zhou et al. 2012 and Zhou et al. 2013).

The third step is to construct BDS-3, expanding the coverage from the Asia–Pacific region to the globe (CSNO 2019a). The development of BDS-3 started in 2009. On November 5, 2017, the first two BDS-3 satellites were deployed, marking the first step of BDS-3 construction. By December 27, 2018, 18 MEO satellites and 1 GEO satellite had been put into the sky. The completion of the BDS-3 basic system and the opening up of global PNT services were announced by the China Satellite Navigation Office on that day.

On June 23, 2020, the last BDS-3 GEO satellite was successfully launched, indicating that the deployment of the BDS-3 constellation was finished. The BDS-3 constellation includes 24 MEO satellites, 3 IGSO satellites and 3 GEO satellites. On July 31, 2020, the Chinese government officially announced the commissioning of BDS-3. Compared to BDS-2, the broadcast orbit and clock accuracy of BDS-3 satellites were considerably improved (Tang et al. 2016; Chen et al. 2020). As reported by Chen et al. (2020), the signal-in-space accuracy of BDS-3 is better than 0.5 m, the positioning accuracy is better than 10 m, and the timing accuracy is better than 20 ns. BDS-3 provides three kinds of services to users world-wide and four kinds of services to users around China. The world-wide services include global PNT services, global short message communication (GSMC) service and international search and rescue (SAR) service. The regional services include satellite-based augmentation system (SBAS), ground-based augmentation system (GBAS), precise point positioning (PPP) and regional short message communication (RSMC) services (CSNO 2019a).

PPP enables a single receiver to achieve centimetre-level to decimetre-level positioning accuracy in either static mode

or kinematic mode and is regarded as a popular and widely used technology in high-precision positioning applications of GNSS. Due to its high accuracy, PPP is widely used in crustal deformation monitoring, near real-time GPS meteorology, orbit determination of low Earth orbiting satellites for gravity recovery missions and the precise positioning of mobile objects (Calais et al. 2006; Rocken et al. 2005; Zhu et al. 2004 and Zhang and Andersen 2006). Precise orbit and clock products with centimetre-level accuracy are the prerequisites of PPP services (Zumberge et al. 1997 and Kouba and Héroux 2001).

PPP is one of the featured services of BDS-3 for supporting high-precision positioning applications (Liu et al. 2020). The PPP-B2b signal, transmitted by BDS-3 GEO satellites, broadcasts precise orbit corrections, clock corrections and Differential Code Biases (DCB) of BDS-3 IGSO/MEO satellites and GPS satellites to dual-frequency users around China and its surrounding areas (CSNO 2019b). Currently, the supported GNSS constellations include BDS-3 IGSO/MEO satellites and GPS satellites. The precise corrections enable GPS/BDS dual-frequency PPP. In this contribution, the generation strategy of precise corrections for PPP-B2b service is briefly introduced as well as accuracy evaluation.

The paper is organized as follows. The PPP-B2b broadcast messages and information types are briefly introduced in Sect. 2. The orbit determination and clock estimation strategy with accuracy assessment are given in Sect. 3. The positioning accuracy for a PPP-B2b broadcast message is evaluated in Sect. 4. Finally, the conclusions and discussion are given in Sect. 5.

2 PPP-B2b broadcast message and information type

The Interface Control Document (ICD) of PPP-B2b was released by the China Satellite Navigation Office at the end of 2019. According to the ICD, the PPP-B2b signal broadcasts the I-component and the Q-component. Currently, the first three GEO satellites of BDS-3 broadcast only the I-component. The broadcast PPP-B2b messages are contained in the I-component. The Q-component will be discussed in the future (CSNO 2019b). The PPP-B2b ICD defines seven information types transmitted in the PPP-B2b I-component. Currently, the PPP-B2b signal broadcasts four of the seven information types. Same as the four information types, the other three information types are also used to broadcast the orbit corrections, clock corrections and integrity information with less communication bandwidth. By using the four information types, the GPS or BDS-3 users are able to get a PPP solution. The current regularly broadcasted information types are given in Table 1.

Table 1 Regularly broadcasted PPP-B2b information types

Type No.	Type name	Update interval (s)	Nominal validity (s)
Type 1	PRN mask	48	–
Type 2	Orbit correction and user range accuracy index	48	96
Type 3	Differential code bias	48	86,400
Type 4	Clock correction	6	12

The PPP-B2b broadcasted information types include several slow variables and a fast variable. The slow variables include the satellite mask, orbit correction and user range accuracy index, and differential code bias, which are broadcasted every 48 s. The fast variable is clock correction, which is broadcasted every 6 s. Note that "Nominal validity" in Table 1 gives the time interval of validity of the message. For message outside this interval the data quality cannot be ensured.

With the information types mentioned in Table 1 and the navigation messages broadcasted by BDS-3 IGSO/MEO satellites and GPS satellites, precise orbits and clock offsets can be recovered.

1) *PRN mask*

The PRN mask is to define which satellites are used for PPP services so that each message type can share a common set of satellites defined in advance, thereby saving the message bits. The PRN mask defines whether the precise corrections of a specific satellite are broadcasted. If the precise corrections of a specific satellite will be broadcasted, the corresponding bit in the mask will be assigned "1". Satellite orbit correction, clock correction and differential code bias are used matching the PRN mask. PRN mask is the index table which is used to assign corrections to the satellites. Satellite masks are usually kept fixed; otherwise, a new satellite joins the PPP service or a retired satellite is taken away. The IODP (Issue Of Data, PRN mask) defines the versions of the satellite mask. IODP is usually kept fixed unless the bits in PRN mask are changed. IODP also exists in the orbit corrections (Type 2) and differential code biases (Type 3). PRN mask, orbit corrections and differential code biases with different IODP values should not be used together.

2) *Orbit correction parameters and user range accuracy index*

Orbit corrections are expressed in terms of radial, along-track and cross-track corrections through Type 2 (Orbit

Correction Parameters and User Range Accuracy Index). Note that the orbit corrections of BDS satellites refer to the CNAV-1 message broadcasted by B1C (CSNO 2017) and that the orbit corrections of GPS satellites refer to the L-NAV of GPS L1 and L2 (GPS Directorate 2019). Using the orbit corrections and the broadcasted navigation message, users can obtain the satellite orbits relative to the satellite phase centre. Therefore, no satellite phase centre offsets corrections should be applied by the users. The term IODN (Issue Of Data, Navigation) is used to match Type 2 and the broadcasted navigation message. IODN of Type 2 is expected to be equal to IODC (Issue Of Data, Clock) in a CNAV-1 message or in an L-NAV message. The orbit correction parameters and user range accuracy index are used only if IODPs in the type and those in the PRN mask are equal.

3) *Differential Code Bias*

The differential code bias defines the transmitting delay difference of the signal components involved with respect to the satellite clock parameters. Currently, the B1C, B2a, B2b, B1I and B3I differential code bias parameters are broadcasted. The differential code bias corrections enable multiple ionosphere-free linear combinations to be available to users. The differential code bias should be used in matching the PRN mask by the IODP.

4) *Clock correction parameters*

Clock correction parameters also refer to CNAV-1 messages broadcasted by B1C and L-NAV of GPS L1 and L2. The clock correction parameters should be used matching the orbit corrections with the same IOD Cor. A clock correction is invalid if it equals -26.2128 m. If a clock correction equals -26.2128 m, users should not use this satellite for PPP.

Estimation of clock corrections is aligned with BDT. However, the PPP-B2b ICD defines that the absolute value of clock corrections should be no larger than 26.2128 m, which is not large enough to contain the difference between GPST and BDT. Furthermore, no accurate GPST/BDT difference parameters are broadcasted by PPP-B2b. Therefore, for BDS/GPS dual-system users, two different receiver clock offsets should be set and estimated epoch by epoch. If the dual-system users estimate one receiver clock offset epoch by epoch and one constant inter-system bias parameter, the positioning accuracy may get worse.

3 Orbit and clock correction generation and accuracy assessment

This section concentrates on the orbit and clock correction accuracy of PPP-B2b broadcast message. A brief description of the satellite orbit and clock correction generation process

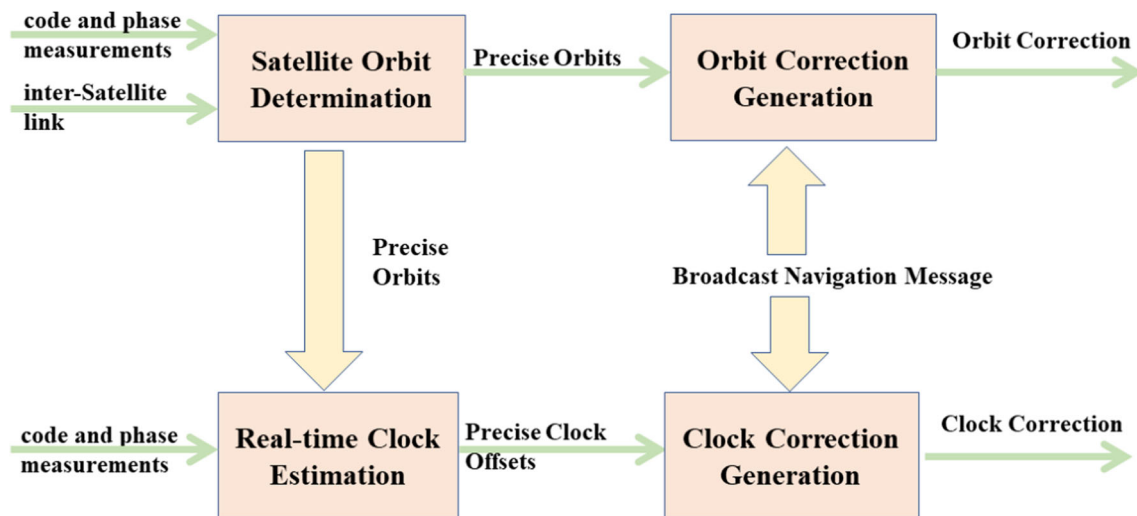


Fig. 1 Main flow chart of satellite orbit and clock correction generation

is presented, followed by the accuracy assessment of the orbit and clock corrections of GPS and BDS-3 satellites.

3.1 Satellite orbit and clock correction generation

Precise satellite orbit and clock correction generation for PPP-B2b is conducted using the orbit and clock determination software developed by Shanghai Astronomical Observatory, Chinese Academy of Science and Beijing Satellite Navigation Centre. The software is operated at the control segment. Satellite orbit and clock correction determination is divided into four different processes, as shown in Fig. 1.

The satellite orbit determination process is conducted periodically in batch mode and estimates precise BDS and GPS orbit solutions using code/phase measurements from ground monitoring receivers and inter-satellite link measurements in combination. Orbit corrections are generated by differencing precise orbit solutions with broadcast orbits in the orbit correction generation process.

In consideration of the irregular behaviour and random variations of the on-board atomic clock, the real-time clock estimation process takes code and phase measurements as input and estimates satellite clock offsets using code and phase measurements in real-time with a Kalman filter. To reduce the computation burden and time loss, the satellite orbits are no longer estimated but fixed to solutions generated by the satellite orbit determination process. Finally, clock corrections are generated by differencing precise clock offset solutions with broadcast clock parameters in the clock correction generation process.

3.1.1 Satellite orbit determination strategy

The satellite orbit determination process follows the statistical orbit determination form using the least-squares

estimation (Weiss et al. 2017). Precise orbits are usually generated by the orbit determination process using code and phase measurements from monitoring stations within a global tracking network. A new orbit determination strategy of BDS-3 satellites is implemented for the PPP service, in which the BDS-3 orbits are estimated using inter-satellite link (ISL) measurements and observations from 7 stations in mainland China in combination instead of the globally-distributed monitoring stations. By doing so, maintenance cost of the globally-distributed stations is saved. As indicated from Yang et al. (2019a), inter-satellite link measurements of BDS-3 follow a time division multiple access (TDMA) structure. According to Tang et al. (2018), a pair of satellite A and B in view receives pseudorange measurements from each other at different times. The observation equations are,

$$\begin{aligned}
 \rho_{AB}(t_1) &= \left| \vec{R}_B(t_1) - \vec{R}_A(t_1 - \Delta t_1) \right| + c * clk_B(t_1) \\
 &\quad - c * clk_A(t_1) + c * \tau_A^{Send} + c * \tau_B^{Rcv} + \Delta \rho_{cor}^{AB} \\
 \rho_{BA}(t_2) &= \left| \vec{R}_A(t_2) - \vec{R}_B(t_2 - \Delta t_2) \right| \\
 &\quad + c * clk_A(t_2) - c * clk_B(t_2) + c * \tau_B^{Send} \\
 &\quad + c * \tau_A^{Rcv} + \Delta \rho_{cor}^{BA}
 \end{aligned} \quad (1)$$

where $\vec{R}_A$ and $\vec{R}_B$ are the vectors representing Cartesian coordinates of satellite A and satellite B, clk_A and clk_B are the clock offsets, Δt_1 and Δt_2 are the signal travel times. τ_A^{Send} and τ_A^{Rcv} are the sending hardware delay and receiving hardware delay for satellite A respectively, τ_B^{Send} and τ_B^{Rcv} are the sending hardware delay and receiving hardware delay for satellite B respectively. $\Delta \rho_{cor}^{AB}$ and $\Delta \rho_{cor}^{BA}$ are the easily-modelled errors, such as phase centre offset corrections, gravitational time delay and relativistic effect due to orbital eccentricity.

Using predicted satellite orbit and clock parameters, the geometrical distances can be derived with accuracy loss of less than 1 cm:

$$\frac{\rho_{AB}(t_0) + \rho_{BA}(t_0)}{2} = \left| \vec{R}_B(t_0) - \vec{R}_A(t_0) \right| + c * X_{Delay}^A + c * X_{Delay}^B + \frac{\Delta\rho_{cor}^{AB} + \Delta\rho_{cor}^{BA}}{2} \quad (2)$$

where, $X_{Delay}^A = \frac{\tau_A^{Send} + \tau_A^{Rcv}}{2}$, $X_{Delay}^B = \frac{\tau_B^{Send} + \tau_B^{Rcv}}{2}$, t_0 is the target time which is usually chosen as t_1 or t_2 .

The derived geometrical distances between the two satellites are regarded as inputs in the orbit determination together with observations from the regional network. The ISL measurements have advantages for orbit determination because of the geometrical distances that are free of clock offsets and wider coverage. Furthermore, while the ground stations track the satellites nearly parallel with the radial direction, the ISL measurements are more sensitive to along-track and cross-track directions. By geometrical distances and regional observations in combination, the orbit solutions are accurate and comparable to those from 30 globally-distributed monitoring receivers (Yang et al. 2019b).

However, the precise orbit determination strategy of GPS satellites is different from that of BDS-3 satellites. GPS orbit determination and BDS-3 orbit determination are conducted separately in two processes. For GPS satellites, un-differenced L1/L2 ionosphere-free code and phase combinations from 30 globally-distributed stations are used in orbit determination.

The satellite orbit determination process is conducted every 1 h with a 3-day arc with sampling interval of 60 s. Then one-hour predicted orbit is generated using the estimated dynamical variables. The orbit corrections are generated using the predicted orbit.

The detailed orbit determination strategy is summarized in Table 2. In the orbit determination process, float ambiguities are used. The ECOM solar radiation model is used without prior model and five acceleration parameters are estimated including constant accelerations in D, B and Y directions, cosine and sine accelerations in the B direction during a 3-day arc (Springer et al. 1999). During eclipses, the eclipsing factor is modelled in the solar radiation model and the attitude angles are modelled following Kouba (2009) and Montenbruck et al. (2015) for GPS satellites and laws provided by the manufacturers for BDS-3 satellites (CSNO 2019c).

From Eq. (2), the unknown hardware delays are present in the geometrical distances derived from ISL dual one-way measurements. Therefore, one hardware delay parameter per satellite is set and estimated. The estimated parameters include 6 initial satellite elements and 5 solar pressure model parameters per satellite, one station-specific ZTD parameter

every 2 h, float ambiguities, satellite-specific ISL hardware delays and epoch-wise clock offsets. The receiver and satellite clock offsets are treated as epoch-wise white noise in orbit determination. For BDS-3 orbit determination with 3-day arc and 60 s sampling, 34 clock offsets per epoch are estimated including 27 BDS-3 satellites and 7 receivers. The total number of clock offsets reaches more than 70,000. If other global variables are estimated together with the epoch-wise clock offsets, the computation burden will become too big. Therefore, the clock offsets elimination method by Zhou et al. (2011) and Tang et al. (2016) is applied.

3.1.2 Real-time satellite clock estimation strategy

The satellite clock estimation process is conducted in real-time with a Kalman filter using un-differenced ionosphere-free code and phase combinations from the regional network within mainland China by fixing predicted orbits from the satellite orbit determination process. The clock offsets of satellites and receivers are modelled as white noises and are estimated epoch-wise. To avoid rank deficiency, the offset of one specific clock should be regarded as the clock datum and kept fixed on known values. In the proposed strategy, clock offsets of the receiver located at the master station and aligned to BDT are kept as the clock datum to estimate offsets of the satellites and other ground receivers. By doing so, the estimated precise satellite clock offsets refer to BDT. Due to the possible irregular behaviour of the clocks, the clock offsets of satellites and receivers are estimated as white noise and the process noise is $1 \text{ ms}/\sqrt{1\text{s}}$.

The tropospheric delay is corrected with the Saastamoinen model and GMF (Boehm et al. 2006; Saastamoinen 1972). The residual tropospheric delay is modelled and estimated as a random-walk process. The process noise is $1 \text{ cm}/\sqrt{1\text{hour}}$. The float ambiguities are also estimated. A detailed real-time satellite clock estimation strategy is summarized in Table 3.

3.2 Accuracy of orbit and clock corrections of PPP-B2b broadcast message

The accuracy of the PPP-B2b broadcast message is evaluated by comparing the orbit and clock corrections with the MGEX (multi-GNSS) precise products generated by the GeoForschungsZentrum Potsdam (GFZ) analysis centre (Montenbruck et al. 2013, 2017; Steigenberger and Montenbruck 2020).

3.2.1 Accuracy of orbit solutions

Evaluation is done using PPP-B2b broadcast messages from August 1, 2020, to August 7, 2020. The 3D orbit accuracy of the MGEX GFZ products is 5 cm for GPS by orbit comparisons (Steigenberger and Montenbruck 2020 and Kazmierski

Table 2 Strategy of the orbit determination process

Orbit determination process strategy	
Observations	For BDS-3 satellites, un-differenced B1I/B3I ionosphere-free code and phase combinations from a regional network containing 12 stations within mainland China and the geometrical distances between two satellites derived from ISL measurements are used in combination For GPS satellites, un-differenced L1/L2 ionosphere-free code and phase combinations from 30 globally-distributed stations
Elevation angle cut-off	7°
Arc length	3 days
Sampling	60 s
Satellite PCO	Values from manufacturers for BDS-3, igs14.atx for GPS (Rebischung et al. 2016)
Receiver antenna PCO	Values from manufacturers. The receiver antennas used are made by Chinese manufacturers, whose PCO values are not provided by igs14.atx
Satellite PCV	Not applied for BDS-3, igs14.atx for GPS (Rebischung et al. 2016)
Receiver antenna PCV	Not corrected. The receiver antennas used are made by Chinese manufacturers, whose PCV values are provided by neither igs14.atx nor manufacturers
Station coordinates	The station coordinates are fixed in the BDCS frame with is aligned to ITRF2014, which are kept as known values in orbit determination
Troposphere delay	Saastamoinen model (Saastamoinen 1972) for wet and dry hydrostatic delay with the GMF (Boehm et al. 2006) without the gradient model
Ionosphere delay	The first-order ionosphere delay is eliminated by the dual-frequency combinations
Orbit models	Two-body motion and non-spherical Earth; the attracting forces of the sun and the moon; tidal potential and solar radiation pressure
Solar radiation model	Constant accelerations in the D direction, B direction and Y direction; cosine and sine accelerations in the B direction (Springer et al. 1999). D is the direction Satellite-Sun. Y points along the satellite's solar panels axes, and B completes the orthogonal system
Estimated parameters	The initial satellite positions and velocities, 5 solar pressure model parameters, satellite and receiver clock offsets, one zenith tropospheric delay parameter (ZTD) per hour and carrier phase ambiguities, satellite-specific ISL hardware delays
Estimation	Least-squares estimation

et al. 2018). Currently, the accuracy of BDS-3 orbits has not yet been estimated. The 3D accuracy is 18 cm for BDS-2 MEO satellites and 36 cm for BDS-2 IGSO satellites (Steigenberger and Montenbruck 2020 and Kazmierski et al. 2018).

Note that the PPP-B2b orbits and MGEX GFZ orbits refer to different points. The PPP-B2b BDS-3 orbits are relative to the satellite B3 phase centre, the PPP-B2b GPS orbits are relative to the L1/L2 ionosphere-free combination phase centre of the satellite, and the MGEX GFZ orbits are relative to the satellite centre-of-mass. The satellite phase centre offsets are corrected in the comparisons. The orbit differences are used as follows:

$$\vec{e} = \vec{B} - \mathbf{Tr} * \vec{P} - \vec{R} \quad (3)$$

In the above equation, $\vec{e}$ refers to the orbit difference, $\vec{B}$ means the broadcasted orbits of the navigation message, $\vec{P}$ refers to the satellite phase centre offsets in the satellite-fixed system, $\mathbf{Tr}$ means the transformation matrix from the satellite-fixed system to the Earth-fixed system, which can be

calculated by the attitude models reported by Kouba (2009) and Montenbruck et al. (2015) for GPS satellites and provided by the manufacturers for BDS-3 satellites (CSNO 2019c). $\vec{R}$ refers to the precise orbits of the MGEX GFZ. The phase centre offset values from the IGS-ANTEX file are used for the GPS and BDS-3 orbit comparisons.

The root mean square (RMS) values of the orbit differences are given in Fig. 2 for the BDS and in Fig. 3 for the GPS. The mean RMS values by satellite type are summarized in Table 4. The user ranging error (URE) is also calculated and given in the figures and table. The URE is calculated as follows (Zhou et al. 2011):

$$\text{For IGSO, } URE = \sqrt{R^2 + (0.09 * A)^2 + (0.09 * C)^2} \quad (4)$$

$$\text{For MEO, } URE = \sqrt{(0.99 * R)^2 + (0.14 * A)^2 + (0.14 * C)^2} \quad (5)$$

where R, A and C refer to the orbit error in the radial, along-track and cross-track directions, respectively.

Table 3 Strategy of the real-time satellite clock estimation process

	Real-time satellite clock estimation strategy
Observations	Un-differenced ionosphere-free code and phase combination from a regional network containing 12 stations within Mainland China
Frequency band	L1/L2 ionosphere-free code and phase combinations for GPS; B1I/B3I ionosphere-free code and phase combinations for BDS-3
Sampling	6 s
Elevation angle cut-off	7°
Satellite orbits	Phase centre orbits are obtained by adding phase centre offsets to centre-of-mass predicted orbits; phase centre orbits are fixed
Observation weight	Code: phase = 1:10,000
Receiver antenna PCO	Values from manufacturers. The receiver antennas used are made by Chinese manufacturers, whose PCO values are not provided by igs14.atx
Satellite PCV	Not applied for BDS-3; igs14.atx for GPS (Rebischung et al. 2016)
Receiver antenna PCV	Not corrected. The receiver antennas used are made by Chinese manufacturers, whose PCV values are provided by neither igs14.atx nor manufacturers
Station displacement	Solid Earth tide, pole tide, ocean tide loading corrected (Petit and Luzum 2010)
Relativity	Corrected (Kouba 2004)
Troposphere delay	Saastamoinen model (Saastamoinen 1972) for wet and dry hydrostatic delay with the GMF (Boehm et al. 2006) without the gradient model; residual tropospheric delay as a random-walk process. The process noise of the residual tropospheric delay is $1 \text{ cm}/\sqrt{1 \text{ hour}}$
Ionosphere delay	The first-order ionosphere delay is eliminated by the dual-frequency combinations
Receiver clock	Estimated as white noise, the initial value is given as 0 and prior covariance is given as $9 \times 10^{12} \text{ m}^2$. Clock offsets of the receiver located at the master station and aligned to BDT are kept as the clock datum to estimate offsets of the satellites and other ground receivers
Satellite clock	Estimated as white noise, the initial value is given as 0 and prior covariance is given as $9 \times 10^{12} \text{ m}^2$
Phase ambiguities	Estimated as constant parameters
Inter-System Biases	Estimated as random-walk variables
Estimation	Kalman filter

The orbit differences of BDS-3 IGSO satellites (C38, C39 and C40 in Fig. 2) are larger than those of BDS-3 MEO satellites and GPS satellites. The mean radial orbit error is approximately 0.155 m, the 3D orbit error is approximately 0.328 m, and the URE is approximately 0.157 m for the BDS-3 IGSO satellites. The orbit differences of the BDS-3 MEO satellites are at the same level as those of the GPS satellites. The radial orbit differences are approximately 0.05 m, the 3D orbit differences are approximately 0.20 m, and the orbit URE is better than 0.06 m.

From Fig. 2, it can be clearly seen that the radial errors of C19-C22 and C41-C46 seem to be larger than those of other MEO satellites. The available ISL measurements may account for the phenomenon. The available ISL measurements for the four satellites are fewer than other MEO satellites by 15% during that period.

The ISL measurements are insensitive to the orbital orientation parameters including inclination angle and right ascension of ascending intersection. In the ISL enhanced BDS-3 orbit determination, the reduction of orbit error in the cross-track is smaller than in the radial and along-track

directions, resulting in larger orbit errors in the cross-track direction.

3.2.2 Accuracy of clock solutions

The MGEX GFZ precise clock is taken as the reference for evaluating the accuracy of the PPP-B2b clock solutions. The PPP-B2b broadcast message with the same period as that of the orbit evaluation is used for the clock accuracy evaluation. Both the broadcasted PPP-B2b GPS clock corrections and MGEX GFZ precise GPS clock offsets refer to the L1P/L2P ionosphere-free combinations. However, the GFZ precise BDS-3 clock offsets refer to the B1I/B3I ionosphere-free combinations, while the broadcasted PPP-B2b BDS clock corrections refer to B3I. Therefore, the MGEX DCB parameters provided by the Chinese Academy of Sciences (CAS) are used to correct the PPP-B2b BDS clock corrections before comparison (Wang et al. 2016).

Following Yao et al. (2017), the accuracy assessment of the clock solutions follows a two-step process. Firstly, the PPP-B2b satellite clock offsets are differenced with the GFZ

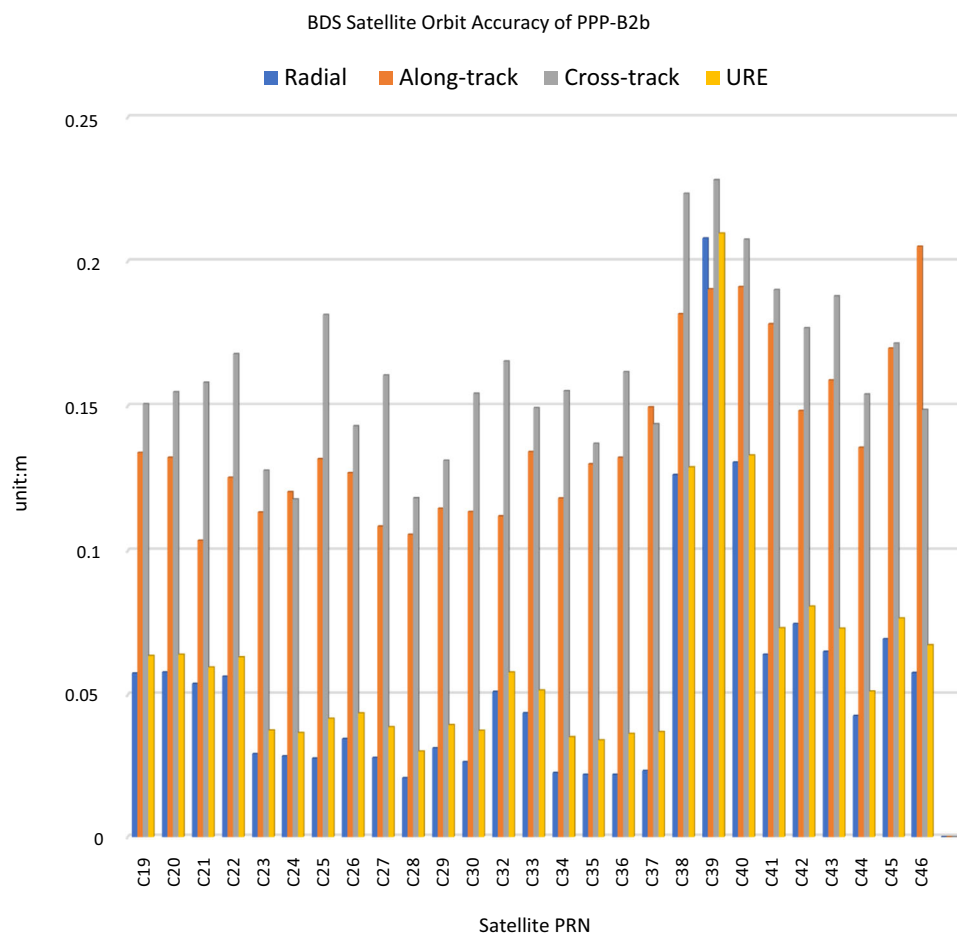
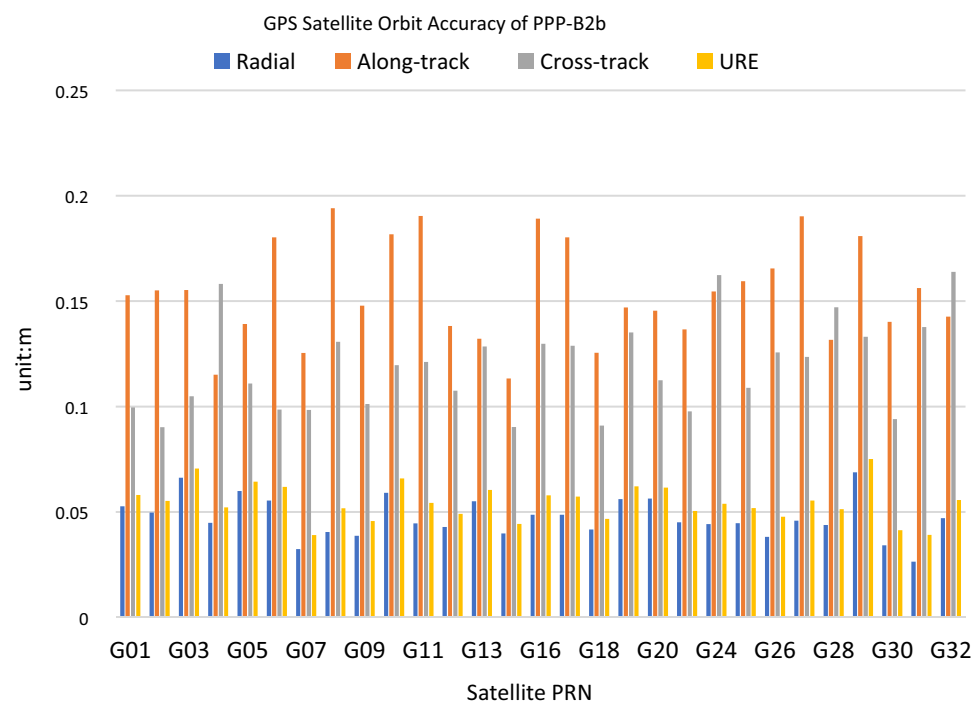
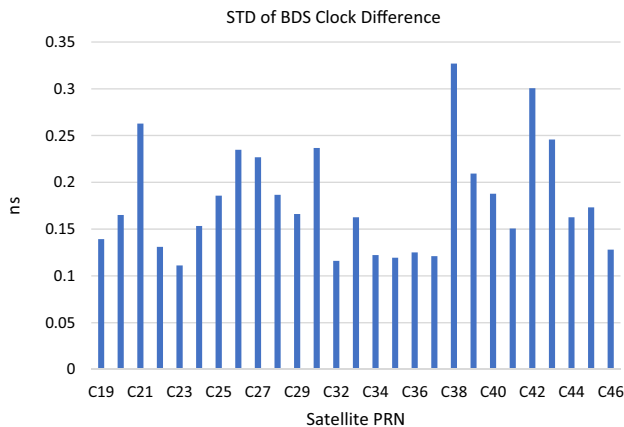
Fig. 2 BDS satellite orbit accuracy of PPP-B2b**Fig. 3** GPS satellite orbit accuracy of PPP-B2b

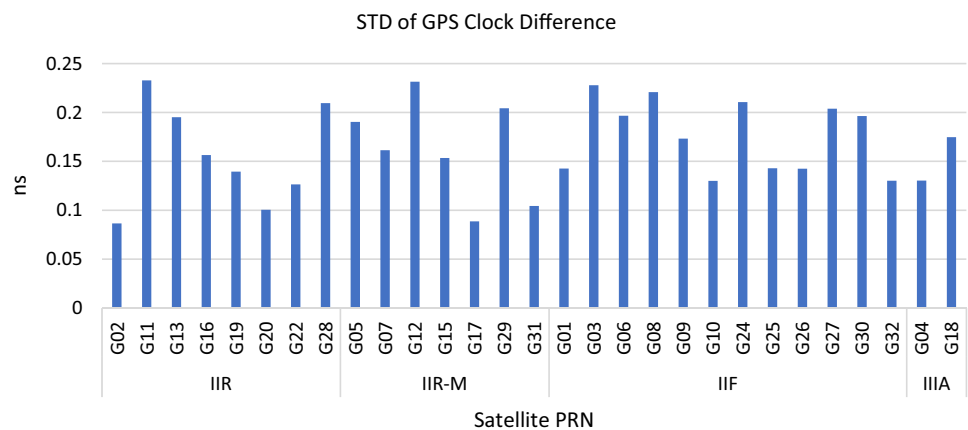
Table 4 RMS of orbit and clock differences by satellite type

Satellite type	Orbit (m)					STD of clock difference (ns)
	R	T	N	3D	URE	
BDS-3 IGSO	0.155	0.188	0.220	0.328	0.157	0.241
BDS-3 MEO	0.042	0.133	0.155	0.210	0.051	0.172
GPS block IIR	0.050	0.153	0.120	0.202	0.057	0.156
GPS block IIR-M	0.046	0.148	0.115	0.193	0.053	0.162
GPS block IIF	0.047	0.164	0.119	0.209	0.055	0.177
GPS block IIIA	0.043	0.120	0.125	0.181	0.049	0.153

**Fig. 4** STD of BDS-3 Satellite Clock Differences

precise clock offsets, Secondly, the mean clock differences for all available satellites are subtracted from the above clock differences to remove differences of time reference. For satellite i , the clock differences can be written as:

$$\Delta clk_i(t) = clk_i^{B2b} - clk_i^{GFZ} - \sum_{n=1}^{ns} \frac{clk_n^{B2b} - clk_n^{GFZ}}{ns} \quad (6)$$

Fig. 5 STD of GPS Satellite Clock Differences

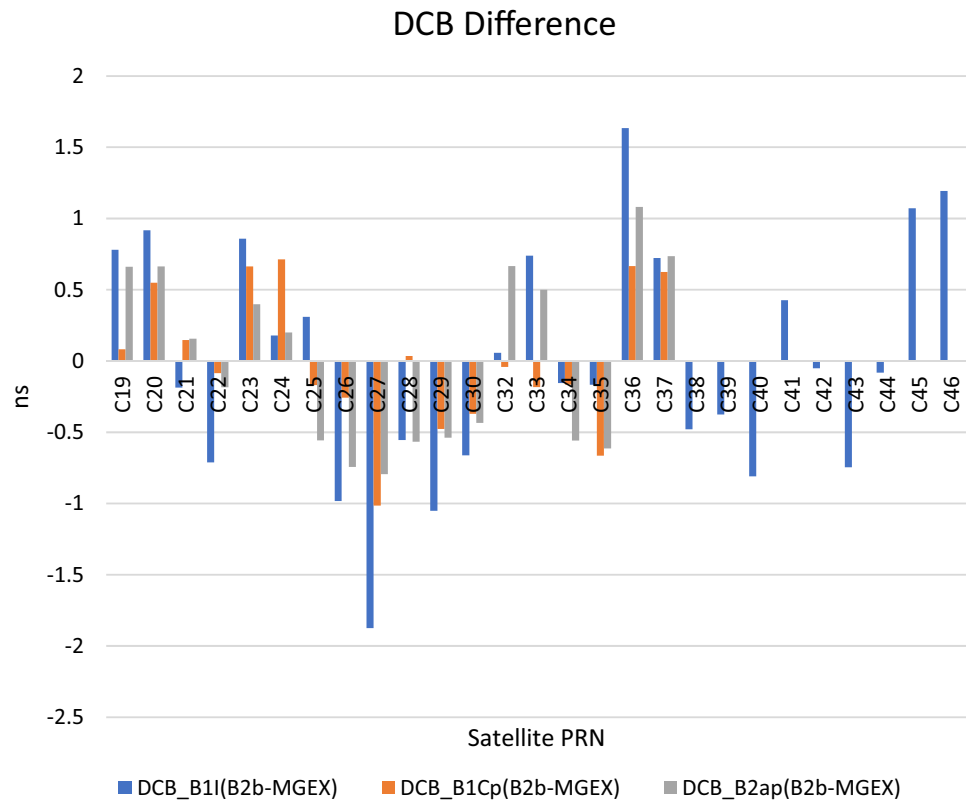
clk_i^{B2b} and clk_i^{GFZ} are the PPP-B2b satellite clock offsets and GFZ precise clocks for satellite i respectively, ns is the number of available satellites.

Standard Deviations (STDs) of the BDS-3 clock differences and GPS clock differences are given in Figs. 4 and 5, respectively. The statistics of the clock differences by satellite type are also given in Table 4. As shown in Table 4 and Fig. 5, the STDs of most GPS satellite clock differences are less than 0.3 ns. The STD of clock differences is not satellite-type dependent for GPS. The average STD of clock differences is approximately 0.2 ns. The STD of the clock differences of most BDS-3 satellites are at the same level as those of GPS and smaller than 0.3 ns.

3.2.3 Accuracy of DCB parameters

DCB parameters of BDS-3 satellites are provided by the PPP-B2b messages, including DCB(B1I), DCB(B1Cp) and DCB(B2ap). Both the BDS-3 broadcast clock parameters and the PPP-B2b precise clock offsets refer to the B3I signal. DCBs mentioned above refer to differential code biases between the corresponding signal and the B3I signal. Specifically, DCB(B1I) refers to the differential code bias between B1I and B3I. DCB(B1Cp) refers to the differential code bias between B1Cp and B3I. DCB(B2ap) refers to the differential code bias between B2ap and B3I. The

Fig. 6 Comparisons of the PPP-B2b DCB parameters with MGEX DCB products



terms DCB(B1I), DCB(B1Cp) and DCB(B2ap) are compared with MGEX DCB products provided by CAS (Wang et al. 2016). DCB(B1I) parameters of all the BDS-3 satellites, DCB(B1Cp) parameters of C19 to C37 and DCB(B2ap) parameters of C19 to C37 are provided by CAS. The comparison is given in Fig. 6. It can be concluded that the DCB(B1I) errors are smaller than 2.0 ns and DCB(B1Cp) and DCB(B2ap) parameters are smaller than 1.0 ns. RMS of DCB(B1I) errors, DCB(B1Cp) errors and DCB(B2ap) errors are about 0.66 ns, 0.38 ns and 0.56 ns respectively.

The GPS DCBs are not provided by the PPP-B2b signals and users are suggested to conduct L1/L2 dual-frequency GPS PPP.

4 Precise point positioning validation

Performance of the PPP-B2b service is assessed by conducting PPP using measurements from six International GNSS Monitoring and Assessment Service (iGMAS, Jiao 2014) stations in mainland China (bjf1, chu1, kun1, sha1, wuh1 and xia1) as mentioned in Fig. 7 during the period from August 1, 2020, to August 7, 2020. Both the static positioning process and simulated kinematic positioning process are conducted for validation. Three processing modes concerning the systems used are also applied, including BDS-3 only, GPS only

and the GPS/BDS-3 dual system. PPP with GFZ MGEX products is also conducted for comparison purposes.

PPP-B2b corrections used in the validation experiments are decoded by a monitoring receiver provided by College of Electronic Science and Engineering, National University of Defence Technology, located in Shanghai. The receiver decodes the PPP-B2b messages from the three GEO satellites. The receiver outputs precise corrections only if PPP-B2b messages from two of the three GEO satellites pass the consistency check. The PPP strategy is summarized as follows.

4.1 Static PPP validation

In this section static precise point positioning is conducted using PPP-B2b precise corrections. Static PPP accuracy is assessed in terms of convergence time, positioning repeatability and differences with IGS position solutions. Static precise point positioning results with GFZ MGEX products are also conducted for comparison purpose.

4.1.1 Convergence

The PPP-B2b message decoded by the receivers was missing during 23:00-01:00 every day due to a data transmission software problem. Therefore, the PPP is reinitialized at 1:00 every day and ends at 23:00. The convergence time of static

Fig. 7 iGMAS Stations for PPP validation**Table 5** PPP strategy

PPP strategy	
Observations	Undifferenced ionosphere-free code and phase combination from iGMAS stations including bjf1, chu1, kun1, sha1, wuh1 and xia1
Frequency band	L1/L2 ionosphere-free code and phase combinations for GPS; B1I/B3I ionosphere-free code and phase combinations for BDS-3
Elevation angle cut-off	7°
Sampling	60 s
Satellite orbits	For PPP-B2b, satellite orbits relative to the phase centre are obtained by adding orbit corrections to the broadcast orbits; for the GFZ products, centre-of-mass orbits are obtained, with phase centre offsets from igs14.atx added to the centre-of-mass orbits
Observation weight	Code: phase = 1:10,000
Receiver antenna PCO	Values from manufacturers. The receiver antennas used are made by Chinese manufacturers, whose PCO values are not provided by igs14.atx
Satellite PCV	Not applied for BDS-3; igs14.atx for GPS (Rebischung et al. 2016)
Receiver antenna PCV	Not corrected. The receiver antennas used are made by Chinese manufacturers, whose PCV values are provided by neither igs14.atx nor manufacturers
Station displacement	Solid Earth tide, pole tide, ocean tide loading corrected (Petit and Luzum 2010)
Relativistic	Corrected (Kouba 2004)
Troposphere delay	Saastamoinen model (Saastamoinen 1972) for wet and dry hydrostatic delay with the GMF (Boehm et al. 2006) without the gradient model; residual tropospheric delay as a random-walk process
Ionosphere delay	The first-order ionosphere delay is eliminated by the dual-frequency combinations
Receiver clock	For BDS-only or GPS-only PPP, one receiver clock offset is set and estimated as white noise; for BDS/GPS dual-mode PPP, two different receiver clocks are set and estimated as white noise. The initial value is given as 0 and prior covariance is given as $9 \times 10^{12} \text{m}^2$
Receiver position	Estimated as white noise for kinematic PPP and as constant values for static PPP
Satellite clock	Fixed at known values from the PPP-B2b message and GFZ products
Phase ambiguities	Estimated as constant parameters
Estimation	Forward Kalman filter

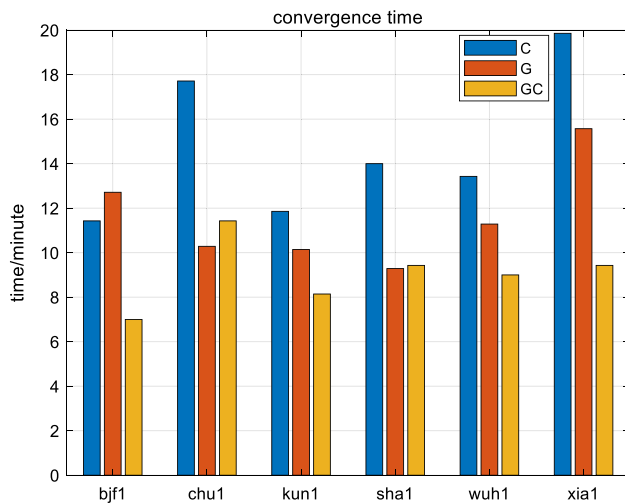


Fig. 8 Convergence time of static PPP

PPP is first evaluated. The convergence condition requires that the horizontal absolute positioning error remains below 20 cm and the vertical remains below 40 cm for at least 5 min for the corresponding direction.

The average convergence times for the six iGMAS stations are given in Fig. 8. Compared to the BDS-3-only or GPS-only PPP, the convergence time of the BDS-3/GPS dual-system PPP is much shorter due to the larger number of satellites and better geometry conditions at most times, except for station chu1, the convergence time of which is a few minutes longer for the BDS-3/GPS dual-system PPP than for the GPS only PPP. Taking xia1 as an example, it requires approximately 20 min to converge for BDS-3-only or GPS-only PPP but converges in less than 10 min for BDS-3/GPS dual-system PPP. For BDS/GPS dual-system static positioning, 7 min to 12 min are required to obtain a convergent positioning result.

4.1.2 Positioning repeatability

The daily repeatability of the static positioning results with the PPP-B2b products is calculated for positioning precision assessment. The daily repeatability with the GFZ products is also calculated for precision comparison purposes. The station-specific repeatability is given in Fig. 9. The average repeatability is summarized in Table 6. From Fig. 9, the repeatability of the positioning results with the GFZ products is slightly better than that with PPP-B2b products in both GPS-only and BDS/GPS dual-system situations. In BDS-only situation, the horizontal positioning repeatability with PPP-B2b is slightly better than that with GFZ products and the vertical positioning repeatability with PPP-B2b is slightly larger than that with GFZ products. The BDS/GPS static positioning repeatability is smaller than 1.0 cm in the north and east directions and smaller than 1.5 cm in the up direction.

From Table 6, the north direction has the smallest repeatability, followed by the east direction. The up direction has the largest repeatability. The average BDS/GPS dual-system positioning repeatability is smaller than 1.0 cm in all directions. The BDS/GPS dual-system positioning repeatability is smaller than that of the BDS-only or GPS-only positioning results in the east and up directions. The differences in the BDS/GPS dual-system positioning repeatability are within 0.5 cm between the PPP-B2b solutions and GFZ solutions.

4.1.3 Differences with IGS position solutions

The average BDS/GPS dual-system daily static positioning results for the above-mentioned period with the GFZ products is taken as the ground truth of the iGMAS station coordinates. Station-specific RMS of the daily positioning errors with PPP-B2b products by taking the GFZ solutions as the ground truth station coordinates as well as the mean RMS are given in Table 7.

Table 7 reveals that the maximum position differences in the north and east directions are 1.47 cm and that the average differences are less than 1.0 cm. However, the position differences in the up direction are much larger. The average RMS of difference in the up direction is approximately 3.0 cm, and the largest differences reach 4.1 cm for wuh1. Larger position differences in the up direction may be attributed to the slightly larger errors of PPP-B2b orbit and clock corrections. From Table 4, orbit errors of PPP-B2b corrections are at decimetre level in the along-track and cross-track directions, which are larger than the MGEX post-processed solutions.

4.2 Simulated Kinematic PPP validation

In this section simulated kinematic precise point positioning is conducted using PPP-B2b precise corrections for iGMAS stations. Although the iGMAS stations are static, the station coordinates are treated as white noise and estimated epoch-wise to assess performance of PPP-B2b service. The accuracy is assessed in terms of convergence time and positioning accuracy after convergence.

4.2.1 Convergence

Simulated kinematic PPP is conducted with the same observations as static PPP. The convergence condition requires that the horizontal absolute positioning error remain below 20 cm and the vertical remain below 40 cm for at least 5 min for the corresponding direction. The average convergence times for the iGMAS stations are given in Fig. 10. From Fig. 10, the convergence time of BDS/GPS dual-system positioning is much shorter than that of the BDS-only or GPS-only positioning solutions. To obtain converged positioning results, 20 min to 51 min are required for BDS-only kinematic PPP,

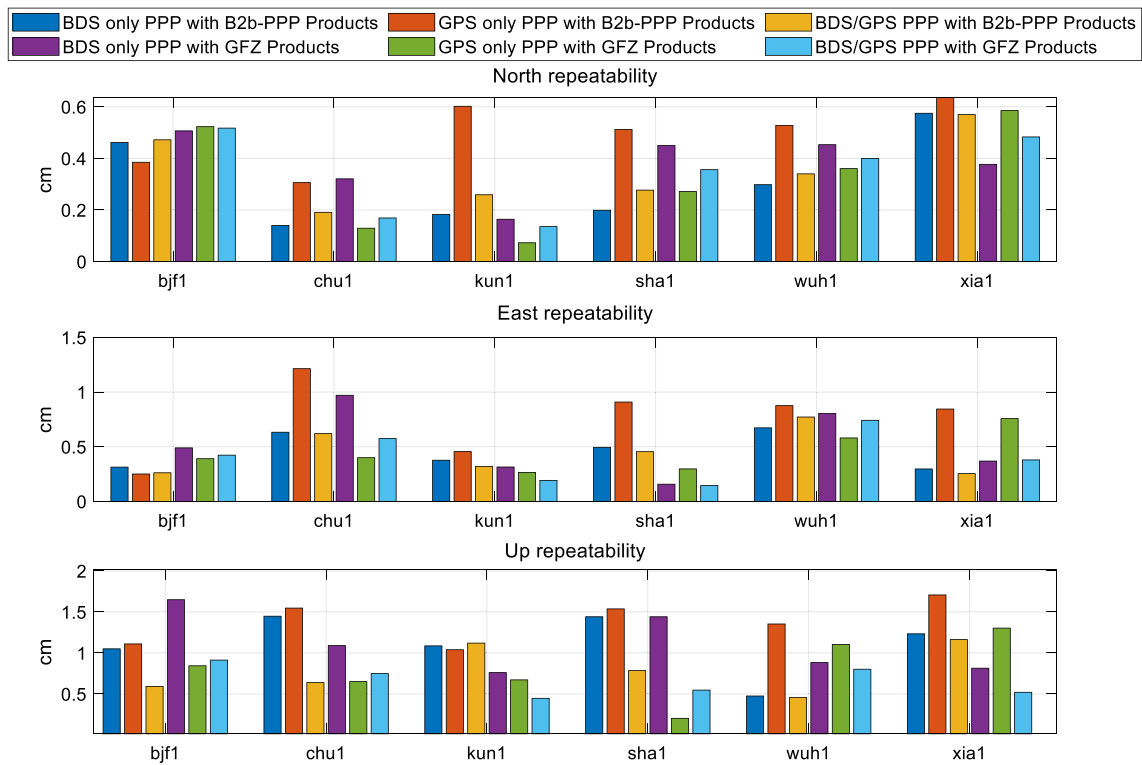


Fig. 9 Daily repeatability of static PPP

Table 6 Average repeatability of static PPP (unit: cm)

Products	<i>N</i>			<i>E</i>			<i>U</i>		
	BDS-only	GPS-only	BDS/GPS	BDS-only	GPS-only	BDS/GPS	BDS-only	GPS-only	BDS/GPS
PPP-B2b	0.309	0.495	0.351	0.465	0.758	0.447	1.122	1.382	0.792
GFZ	0.379	0.324	0.344	0.518	0.449	0.410	0.814	1.302	0.520

Table 7 RMS of difference between PPP-B2b position solutions and GFZ position solutions (unit: cm)

Station	<i>N</i>			<i>E</i>			<i>U</i>		
	BDS-only	GPS- only	BDS/GPS	BDS- only	GPS- only	BDS/GPS	BDS- only	GPS- only	BDS/GPS
Bjf1	0.58	0.39	0.52	1.09	0.53	0.77	2.20	1.92	2.22
Chu1	0.85	0.70	0.79	1.41	1.22	1.14	2.19	1.63	1.22
Kun1	0.88	1.12	0.81	0.95	0.95	0.63	3.41	2.68	3.50
Sha1	0.34	0.62	0.29	0.66	1.26	0.47	3.75	3.85	3.81
Wuh1	0.42	0.60	0.40	1.05	1.47	0.85	3.43	4.10	3.81
Xia1	0.68	0.73	0.60	0.99	0.96	0.76	3.20	3.63	3.39
Mean	0.62	0.69	0.57	1.02	1.07	0.77	3.03	2.97	2.99

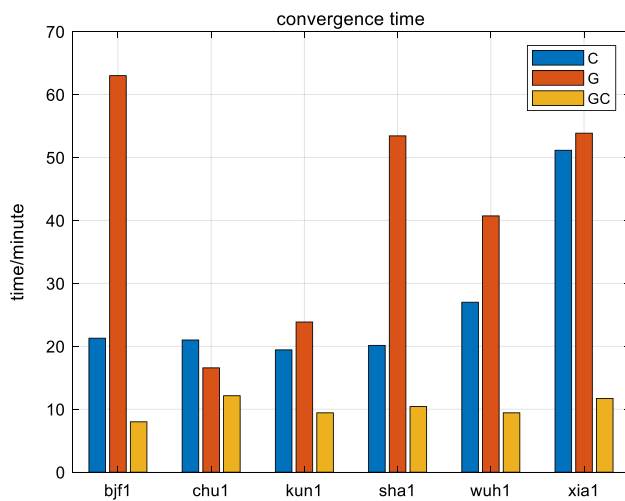


Fig. 10 Convergence time of simulated Kinematic PPP

16 min to 63 min are required for GPS-only kinematic PPP, and less than 12 min are required for GPS/BDS dual-system kinematic PPP.

4.2.2 Positioning accuracy

Average BDS/GPS dual-system daily static positioning results for the above-mentioned period with the GFZ products is taken as the ground truth of the iGMAS station coordinates for simulated kinematic positioning accuracy assessment. The first-hour positioning results after PPP initialization are ignored to avoid the influence of the initialization time on the positioning accuracy evaluation. The simulated kinematic positioning results of the iGMAS station bjf1 are given in Fig. 11. The figure reveals that the GPS/BDS dual-system positioning results are much more stationary and fluctuate less than the GPS-only or BDS-only positioning results.

The RMS of the station-specific simulated kinematic positioning error is given in Fig. 12. As shown, the simulated kinematic positioning error in the north direction is the smallest, followed by that in the east direction, and the up positioning error is the largest. The BDS/GPS dual-system simulated kinematic positioning accuracy is better than the GPS-only or BDS-only simulated kinematic positioning accuracy.

The average RMS values of the simulated kinematic positioning accuracy are listed in Table 8, which shows that the RMS of the BDS-only kinematic positioning errors is 3.2 cm in the north, 5.1 cm in the east and 10.8 cm in the up directions, while it drops to less than 2.2 cm in the north, less than 3.3 cm in the east and less than 8.5 cm in the up directions for BDS/GPS dual-system positioning. The results indicate that the simulated kinematic positioning accuracy obtained using the PPP-B2b real-time precise corrections is at the same level

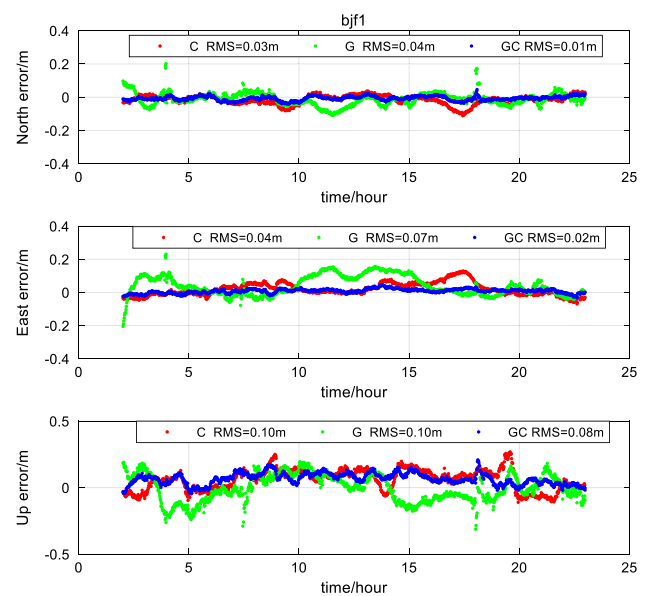


Fig. 11 Simulated Kinematic Positioning Results of bjf1 after Convergence (unit: m) (red or 'C' stands for the BDS-only results, green or 'G' stands for the GPS-only results, and blue or 'GC' stands for the BDS/GPS dual-system results)

as that obtained using the IGS real-time products or QZSS products (Zhang et al. 2018, Wang et al. 2018 and Zhang et al. 2019).

From Table 8, the positioning errors with the PPP-B2b products are from 0.5 cm to 4.0 cm larger than those with the GFZ products for either single-system PPP or dual-system PPP. Remember that the PPP-B2b products are generated in real-time, while the GFZ products are post-processing products (Montenbruck et al. 2013, 2017). Optimal estimations can be obtained in post-processing. However, real-time processing often suffers from quality control and convergence problems, resulting in larger orbits, clock errors and positioning errors (Fu et al. 2019).

5 Conclusions and discussion

PPP is one of the featured services of BDS-3, which provides centimetre-level static positioning accuracy and decimetre-level kinematic positioning accuracy to users around China. The PPP performance is assessed by applying static PPP and kinematic PPP processes to domestic iGMAS stations. Centimetre level static and decimetre level kinematic positioning accuracy can be expected. PPP-B2b service can be used in numerous geodetic fields, such as real-time natural hazards monitoring, water vapour monitoring and precise orbit determination of geodetic LEO satellites.

However, note that the positioning results with the PPP-B2b products are slightly worse than those with the GFZ

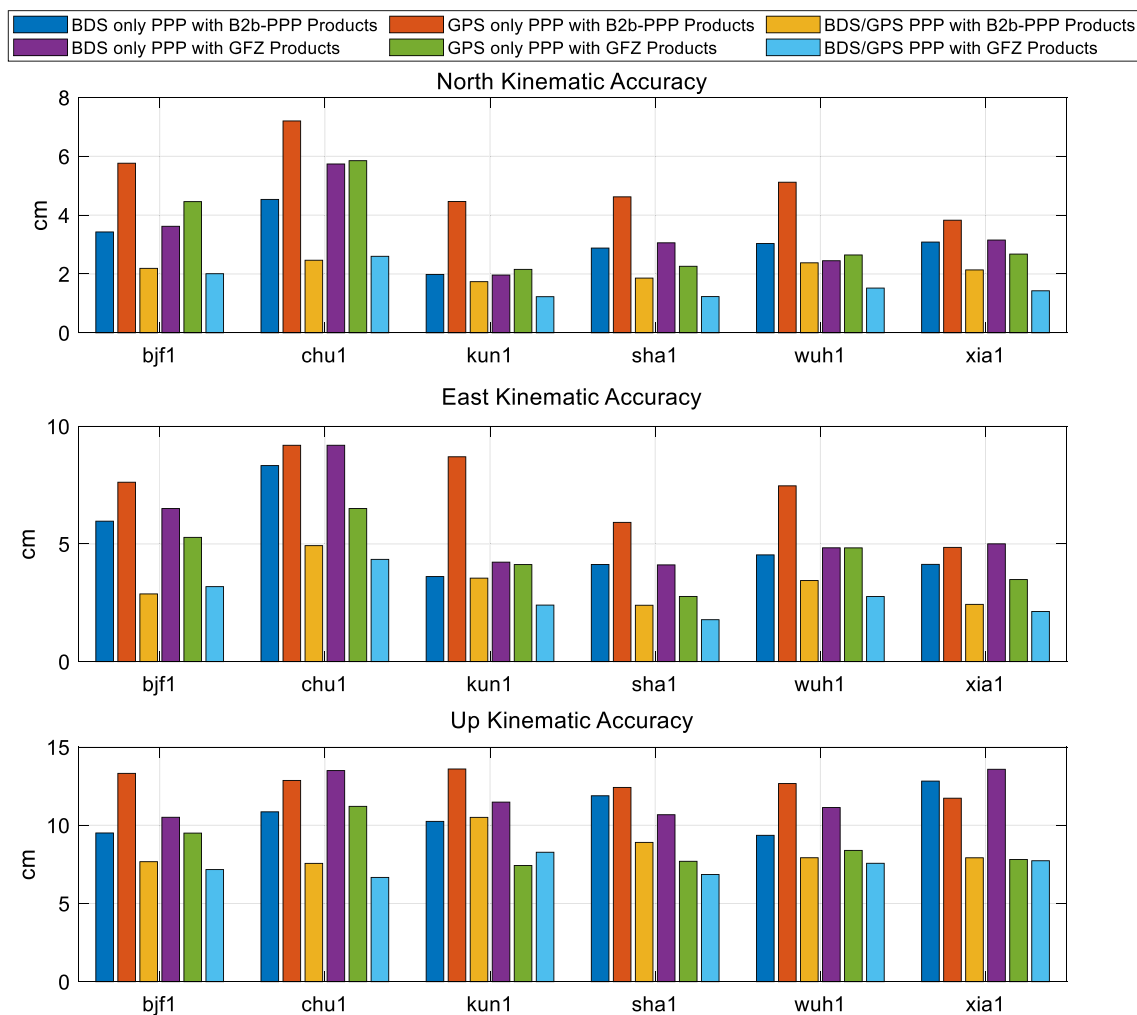


Fig. 12 RMS of positioning accuracy of simulated Kinematic PPP after one-hour initialization

Table 8 RMS of positioning accuracy of simulated Kinematic PPP after one-hour initialization (unit: cm)

Products	N			E			U		
	BDS-only	GPS-only	BDS/GPS	BDS-only	GPS-only	BDS/GPS	BDS-only	GPS-only	BDS/GPS
PPP-B2b	3.157	5.166	2.128	5.118	7.293	3.271	10.778	12.764	8.414
GFZ	3.330	3.341	1.669	5.646	4.499	2.769	11.810	8.672	7.376

post-processed products. The larger orbit error, quality control problems and convergence problems that the real-time process faces may account for the larger positioning errors. The real-time process will be improved for better positioning accuracy in the future, such as addition of more stations, more robust estimation and more accurate phase centre corrections.

Currently, PPP-B2b users require tens of minutes to obtain a converged positioning result. Therefore, further work will concentrate on faster convergence approaches, such as PPP-AR and PPP-RTK. In addition, the PPP integrity

concept, parameterization and monitoring algorithms related to signal-in-space error monitoring and risk alarms will be studied to extend the applicability of PPP-B2b service.

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Author contributions XH and JC provided the initial idea for this study. CT and XH wrote the article. CT and LL developed the real-time clock estimation software and conducted the experiments. SZ and RG conducted the inter-satellite link enhanced orbit determination experiments.

XL and Feng He analysed the information type of PPP-B2b message. Jianhua Yang evaluated orbit and clock accuracy. JL conducted the precise point positioning experiments.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

Data availability The PPP-B2b broadcast message can be obtained with a customized receiver following the public PPP-B2b ICD of BDS-3. Measurements of iGMAS stations for evaluation can be obtained via application. The GFZ products can be download from the IGS website.

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